

Lagrangian Relaxation

General idea is given a LP:

$$\min c^T x$$

$$\text{s.t. } Ax \leq b$$

$$fx \leq d$$

$$x \geq 0$$

We can relax a constraint by removing it from the constraints but adding to the cost:

$$\min c^T x + \lambda(fx - d)$$

$$\text{s.t. } Ax \leq b$$

$$x \geq 0$$

Note that λ references the variable in the dual y_i .